

Brutal Practice Paper B43 - Nutrition in Animals | The Triangle & its Properties | Light | Large Numbers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A cow's small intestine is about 40 m long and a human's is about 6.5 m. By how many metres is the cow's small intestine longer, and why is it so long? [\[Nutrition in Animals\]](#)
 - (A) 46.5 m longer, to digest grass
 - (B) 40 m longer, to absorb sunlight
 - (C) 6.5 m longer, to store water
 - (D) 33.5 m longer, because cows breathe through it
 - (E) 33.5 m shorter, to digest meat
 - (F) 33.5 m longer, to digest large amounts of tough cellulose-rich grass

2. In a triangle, two angles are 47 deg and 68 deg. An exterior angle is formed at the third vertex. What is that exterior angle? [\[The Triangle & its Properties\]](#)
 - (A) 65 deg
 - (B) 115 deg
 - (C) 47 deg
 - (D) 133 deg
 - (E) 85 deg
 - (F) 180 deg
 - (G) 112 deg

3. A plane mirror forms an image as far behind it as the object is in front. If you stand 1.4 m from a mirror, how far is your image from YOU? [\[Light\]](#)
 - (A) 1.4 m
 - (B) 2.0 m
 - (C) 0.7 m
 - (D) 3.5 m
 - (E) 2.8 m
 - (F) 4.2 m
 - (G) 5.6 m

4. Write the number 'twelve crore five lakh seven thousand twenty' in figures (Indian system). [\[Large Numbers\]](#)
 - (A) 1,20,05,07,020
 - (B) 12,05,07,020
 - (C) 1,20,50,7020
 - (D) 12,05,70,020
 - (E) 12,05,07,002
 - (F) 12,05,07,200
 - (G) 12,50,07,020

5. A person has 32 permanent teeth. If the incisors and canines together make up 12 of them, and the premolars number 8, how many molars are there? [\[Nutrition in Animals\]](#)

- (A) 12 molars
- (B) 8 molars
- (C) 20 molars
- (D) 16 molars
- (E) 14 molars
- (F) 4 molars
- (G) 10 molars

6. Two sides of a triangle are 9 cm and 5 cm. The third side must be a whole number of cm. What is the largest possible length of the third side? [\[The Triangle & its Properties\]](#)

- (A) 15 cm
- (B) 5 cm
- (C) 12 cm
- (D) 14 cm
- (E) 9 cm
- (F) 13 cm
- (G) 4 cm

7. Light travels in straight lines. A 2 m tall pole casts a 3 m shadow. At the same time, a tree casts a 12 m shadow. How tall is the tree? [\[Light\]](#)

- (A) 6 m
- (B) 8 m
- (C) 16 m
- (D) 4 m
- (E) 9 m
- (F) 24 m

8. Round 4,67,892 to the nearest ten thousand. [\[Large Numbers\]](#)

- (A) 4,60,000
- (B) 4,70,000
- (C) 4,67,900
- (D) 4,67,000
- (E) 4,68,000
- (F) 5,00,000

9. An amoeba captures food using finger-like extensions and forms a food vacuole. If it engulfs 3 food particles every 2 minutes, how many can it engulf in 18 minutes? [\[Nutrition in Animals\]](#)

- (A) 27 particles
- (B) 6 particles
- (C) 12 particles
- (D) 9 particles
- (E) 30 particles
- (F) 36 particles
- (G) 54 particles

10. A right-angled triangle has legs 6 cm and 8 cm. What is the length of its hypotenuse, and then its perimeter? [\[The Triangle & its Properties\]](#)

- (A) Hypotenuse 10 cm, perimeter 24 cm
- (B) Hypotenuse 10 cm, perimeter 14 cm
- (C) Hypotenuse 10 cm, perimeter 20 cm
- (D) Hypotenuse 7 cm, perimeter 21 cm
- (E) Hypotenuse 48 cm, perimeter 62 cm
- (F) Hypotenuse 14 cm, perimeter 28 cm

11. A concave mirror can form a magnified image; it is used as a shaving/make-up mirror. If it magnifies 3 times, a 2 cm long eyelash appears how long? [\[Light\]](#)

- (A) 6 cm
- (B) 3 cm
- (C) 2 cm
- (D) 0.67 cm
- (E) 12 cm
- (F) 8 cm

12. A factory makes 3,250 toys a day. How many toys does it make in the month of June (30 days), and write the answer in figures. [\[Large Numbers\]](#)

- (A) 97,050
- (B) 90,750
- (C) 975,000
- (D) 32,500
- (E) 97,500
- (F) 98,500

13. Saliva contains an enzyme that begins breaking starch into sugar in the mouth. If chewing bread for 30 seconds breaks down 15% of its starch and you chew twice as long, about how much is broken down (assuming the same rate)? [\[Nutrition in Animals\]](#)

- (A) About 7.5%
- (B) About 60%
- (C) About 50%
- (D) About 45%
- (E) About 75%
- (F) About 30%

14. In an isosceles triangle, each base angle is 52 deg. What is the vertex angle? [\[The Triangle & its Properties\]](#)

- (A) 76 deg
- (B) 26 deg
- (C) 128 deg
- (D) 56 deg
- (E) 38 deg
- (F) 104 deg
- (G) 52 deg

- 15.** White light splits into seven colours through a prism. If a chart shows the colours and you must list how many BOUNDARIES (gaps) lie between adjacent colours, how many are there? [\[Light\]](#)
- (A) 8 boundaries
 - (B) 6 boundaries
 - (C) 1 boundary
 - (D) 5 boundaries
 - (E) 7 boundaries
 - (F) 14 boundaries
 - (G) 3 boundaries
- 16.** Use the comma placement in the International system: write 5,280,000 in words. [\[Large Numbers\]](#)
- (A) Five million twenty-eight thousand
 - (B) Five lakh twenty-eight thousand
 - (C) Five million two hundred eighty thousand
 - (D) Fifty-two million eighty thousand
 - (E) Five million two hundred eight thousand
 - (F) Fifty-two lakh eighty thousand
 - (G) Five hundred twenty-eight thousand
- 17.** In a ruminant, food first enters the rumen, is returned to the mouth as cud, chewed again, and then swallowed for further digestion. A cow brings up 1 cud every 50 seconds. How many cuds can it chew in 15 minutes? [\[Nutrition in Animals\]](#)
- (A) 9 cuds
 - (B) 45 cuds
 - (C) 15 cuds
 - (D) 20 cuds
 - (E) 30 cuds
 - (F) 18 cuds
- 18.** The angles of a triangle are in the ratio 2 : 3 : 4. What is the largest angle? [\[The Triangle & its Properties\]](#)
- (A) 40 deg
 - (B) 60 deg
 - (C) 80 deg
 - (D) 120 deg
 - (E) 90 deg
 - (F) 20 deg
 - (G) 100 deg
- 19.** A convex mirror always makes objects look smaller, giving a wider view. If it shows a 1.8 m car as 0.3 m, by what factor is the image reduced? [\[Light\]](#)
- (A) 5 times smaller
 - (B) 3 times smaller
 - (C) 6 times smaller
 - (D) 1.5 times smaller
 - (E) 0.6 times smaller
 - (F) 9 times smaller
 - (G) 2.1 times smaller

20. What is the difference between the largest 6-digit number and the smallest 6-digit number? [\[Large Numbers\]](#)

- (A) 9,00,000
- (B) 9,99,999
- (C) 8,99,990
- (D) 9,89,999
- (E) 8,99,999
- (F) 1,00,000
- (G) 8,90,999

21. The wall of the small intestine has millions of tiny finger-like villi that increase the surface for absorption. If villi raise the absorbing area to 30 times the flat area, and the flat inner area is 0.5 square metres, what is the absorbing area? [\[Nutrition in Animals\]](#)

- (A) 150 square metres
- (B) 15 square metres
- (C) 1.5 square metres
- (D) 30.5 square metres
- (E) 6 square metres
- (F) 0.6 square metres

22. An equilateral triangle has a perimeter of 27 cm. If a median is drawn, into how many cm is one side split, and what is each whole side? [\[The Triangle & its Properties\]](#)

- (A) Each side 9 cm, median splits it into 3 cm and 6 cm
- (B) Each side 3 cm, halves of 1.5 cm
- (C) Each side 9 cm, halves of 9 cm
- (D) Each side 6 cm, halves of 3 cm
- (E) Each side 13.5 cm, halves of 6.75 cm
- (F) Each side 9 cm, median splits it into two 4.5 cm halves
- (G) Each side 27 cm, halves of 13.5 cm

23. The image in a plane mirror is laterally inverted (left-right swapped). The word displayed on an ambulance reads 'ECNALUBMA' so that in a mirror a driver ahead reads it correctly. How many letters does it have? [\[Light\]](#)

- (A) 8 letters
- (B) 9 letters
- (C) 11 letters
- (D) 9 words
- (E) 6 letters
- (F) 7 letters

24. A library has 1,45,600 books. If 38,750 are issued and 12,400 new books arrive, how many books are now in the library? [\[Large Numbers\]](#)

- (A) 1,18,250
- (B) 1,84,350
- (C) 1,71,750
- (D) 1,19,250 with rounding
- (E) 96,450
- (F) 1,19,250
- (G) 1,06,850

- 25.** During digestion, proteins are broken into amino acids. If a meal contains 24 g of protein and digestion converts $\frac{5}{6}$ of it into absorbable amino acids, how many grams are NOT absorbed? [\[Nutrition in Animals\]](#)
- (A) 4 g
 - (B) 8 g
 - (C) 19 g
 - (D) 5 g
 - (E) 20 g
 - (F) 6 g
- 26.** In a triangle, the first angle is 30 deg more than the second, and the third is 90 deg. What is the second angle? [\[The Triangle & its Properties\]](#)
- (A) 30 deg
 - (B) 45 deg
 - (C) 60 deg
 - (D) 15 deg
 - (E) 120 deg
 - (F) 90 deg
- 27.** Light from the Sun takes about 8 minutes to reach Earth. How many seconds is that, and is light fast or slow on this scale? [\[Light\]](#)
- (A) 8 seconds; light is slow
 - (B) 240 seconds; light is fast
 - (C) 60 seconds; light is fast
 - (D) 4800 seconds; light is fast
 - (E) 80 seconds; light is fast
 - (F) 488 seconds; light is fast
 - (G) 480 seconds; light is extremely fast
- 28.** Estimate the product 612×489 by rounding each number to the nearest hundred. [\[Large Numbers\]](#)
- (A) 3,00,000
 - (B) 60,000
 - (C) 3,06,000
 - (D) 30,00,000
 - (E) 2,99,268
 - (F) 2,40,000
- 29.** The liver makes bile, which is stored in the gall bladder and helps digest fats. If the liver makes 800 mL of bile a day and the gall bladder can hold only $\frac{1}{16}$ of that, what is the gall bladder's capacity? [\[Nutrition in Animals\]](#)
- (A) 100 mL
 - (B) 16 mL
 - (C) 800 mL
 - (D) 40 mL
 - (E) 25 mL
 - (F) 50 mL
 - (G) 80 mL

- 30.** A triangle has sides 7 cm, 24 cm and 25 cm. Is it right-angled, and what is its area if it is? [\[The Triangle & its Properties\]](#)
- (A) Yes, right-angled; area 42 square cm
 - (B) Yes, right-angled; area 300 square cm
 - (C) Yes, right-angled; area 84 square cm
 - (D) Yes, right-angled; area 168 square cm
 - (E) Yes, right-angled; area 600 square cm
 - (F) No, it is not right-angled
 - (G) Yes, right-angled; area 175 square cm
- 31.** A real image can be caught on a screen; a virtual image cannot. A plane mirror always forms a virtual image. Out of 10 mirror reflections you view, how many of these images could be caught on a screen? [\[Light\]](#)
- (A) 0
 - (B) 2
 - (C) 10
 - (D) 4
 - (E) 5
 - (F) 1
 - (G) 9
- 32.** A roll of wire is 1 km long. It is cut into pieces of 250 cm each. How many pieces are obtained? [\[Large Numbers\]](#)
- (A) 100 pieces
 - (B) 4000 pieces
 - (C) 400 pieces
 - (D) 40 pieces
 - (E) 250 pieces
 - (F) 4 pieces
 - (G) 500 pieces
- 33.** Digestion in humans passes through mouth, food pipe, stomach, small intestine and large intestine. If food spends 4 hours in the stomach and three times as long in the small intestine, what is the total time in these two organs? [\[Nutrition in Animals\]](#)
- (A) 24 hours
 - (B) 7 hours
 - (C) 10 hours
 - (D) 8 hours
 - (E) 12 hours
 - (F) 16 hours
 - (G) 20 hours

34. Two angles of a triangle are equal and the third is 96 deg. What is the size of each equal angle? [\[The](#)

[Triangle & its Properties\]](#)

- (A) 84 deg
- (B) 96 deg
- (C) 42 deg
- (D) 21 deg
- (E) 44 deg
- (F) 60 deg
- (G) 48 deg

35. A pinhole camera forms an inverted image. If the object is 30 cm tall and the image is 6 cm tall, what is the magnification (image to object)? [\[Light\]](#)

- (A) 0.5
- (B) $\frac{1}{30}$
- (C) $\frac{1}{5}$ (0.2)
- (D) 6
- (E) $\frac{1}{6}$
- (F) 24

36. Place value: in the number 7,38,46,215, what is the place value of the digit 8? [\[Large Numbers\]](#)

- (A) 80,00,000 (eighty lakh)
- (B) 800 (eight hundred)
- (C) 8,00,000 (eight lakh)
- (D) 80,000 (eighty thousand)
- (E) 8,00,00,000 (eight crore)
- (F) 8,000 (eight thousand)

37. A leech and a mosquito both feed on blood (parasitic nutrition). If a leech can take in 5 times its empty body mass in blood, and its empty mass is 2 g, what is its mass after a full feed? [\[Nutrition in Animals\]](#)

- (A) 5 g
- (B) 12 g
- (C) 15 g
- (D) 2 g
- (E) 11 g
- (F) 7 g

38. An exterior angle of a triangle is 130 deg, and one of its two opposite interior angles is 55 deg. What is the other opposite interior angle? [\[The Triangle & its Properties\]](#)

- (A) 185 deg
- (B) 65 deg
- (C) 85 deg
- (D) 75 deg
- (E) 130 deg
- (F) 55 deg

39. Two plane mirrors placed at right angles (90 deg) form a kaleidoscope-like effect. The number of images formed is $360/\text{angle} - 1$. How many images form at 90 deg? [\[Light\]](#)

- (A) 1 image
- (B) 2 images
- (C) 5 images
- (D) 3 images
- (E) 90 images
- (F) 4 images
- (G) 6 images

40. A stadium holds 65,000 people. For a series of 8 matches that all sell out, how many tickets are sold in total? [\[Large Numbers\]](#)

- (A) 5,20,000
- (B) 5,20,00
- (C) 5,02,000
- (D) 52,000
- (E) 73,000
- (F) 8,000
- (G) 5,200,000

41. Grass-eating animals have flat, broad molars to grind plant matter, while tigers have sharp pointed teeth to tear flesh. A deer grinds 600 g of grass in 30 minutes. At the same rate, how long to grind 1.5 kg?

[\[Nutrition in Animals\]](#)

- (A) 30 minutes
- (B) 50 minutes
- (C) 60 minutes
- (D) 1.5 minutes
- (E) 75 minutes
- (F) 90 minutes
- (G) 45 minutes

42. A ladder leans against a wall. Its foot is 9 m from the wall and it reaches 12 m up the wall. How long is the ladder? [\[The Triangle & its Properties\]](#)

- (A) 225 m
- (B) 15 m
- (C) 10.5 m
- (D) 21 m
- (E) 3 m
- (F) 20 m
- (G) 13 m

43. Sunlight contains seven colours that combine to make white. A spinning disc (Newton's disc) painted in these colours appears white. If it spins 1200 times per minute, how many times per second is that? [\[Light\]](#)

- (A) 72000 times per second
- (B) 12 times per second
- (C) 60 times per second
- (D) 1200 times per second
- (E) 120 times per second
- (F) 20 times per second

44. Arrange in ascending order and name the smallest: 9,09,090; 9,90,009; 9,00,999; 9,09,900. [\[Large Numbers\]](#)

- (A) 9,09,090 ties with 9,00,999
- (B) they are all equal
- (C) 9,90,009 is the smallest
- (D) 9,09,090 is the smallest
- (E) 9,09,900 is the smallest
- (F) 9,00,999 is the smallest
- (G) 9,90,009 is the smallest by ten

45. After absorption, undigested waste collects in the large intestine, where much of the water is absorbed back. If 1.5 L of watery waste enters and 90% of the water is reabsorbed, how much water is reabsorbed?

[\[Nutrition in Animals\]](#)

- (A) 0.9 L
- (B) 1.4 L
- (C) 1.65 L
- (D) 1.5 L
- (E) 0.135 L
- (F) 0.15 L
- (G) 1.35 L

46. In triangle ABC, angle A = $2x$, angle B = $3x$ and angle C = x . Find x and identify the largest angle. [\[The](#)

[Triangle & its Properties\]](#)

- (A) $x = 36$ deg; largest angle is B
- (B) $x = 30$ deg; largest angle is B = 90 deg
- (C) $x = 20$ deg; largest angle is B
- (D) $x = 30$ deg; largest angle is A = 60 deg
- (E) $x = 30$ deg; largest angle is C
- (F) $x = 60$ deg; largest angle is B

47. A man 1.7 m tall wants to see his FULL image in a plane mirror. The shortest mirror needed is half his height. What is the minimum mirror height? [\[Light\]](#)

- (A) 0.6 m
- (B) 1.0 m
- (C) 0.85 m
- (D) 3.4 m
- (E) 0.425 m
- (F) 1.7 m
- (G) 1.2 m

48. How many thousands are there in 4 lakh? [\[Large Numbers\]](#)

- (A) 4,000
- (B) 100
- (C) 40
- (D) 4
- (E) 4,00,000
- (F) 400
- (G) 40,000

49. The tongue helps in tasting, mixing food with saliva and swallowing. Four taste zones detect sweet, sour, salty and bitter. If a test uses 3 samples for each of the 4 basic tastes, how many samples are used in all? [\[Nutrition in Animals\]](#)

- (A) 16 samples
- (B) 3 samples
- (C) 24 samples
- (D) 12 samples
- (E) 4 samples
- (F) 9 samples

50. The two legs of a right triangle are equal, and the hypotenuse is 10 cm. What is the sum of the squares of the two equal legs? [\[The Triangle & its Properties\]](#)

- (A) 1000
- (B) 10
- (C) 25
- (D) 200
- (E) 100 (square cm)
- (F) 20
- (G) 50

51. An object is placed 25 cm in front of a plane mirror and then moved 10 cm closer. What is the new distance between the object and its image? [\[Light\]](#)

- (A) 20 cm
- (B) 25 cm
- (C) 70 cm
- (D) 35 cm
- (E) 30 cm
- (F) 50 cm
- (G) 15 cm

52. A water tank supplies 2,40,000 litres a day to a town. If usage rises by 15,000 litres each day for 4 days, how much is supplied on the 4th extra day (i.e. after 4 increases)? [\[Large Numbers\]](#)

- (A) 2,40,000 litres
- (B) 60,000 litres
- (C) 2,85,000 litres
- (D) 3,60,000 litres
- (E) 3,15,000 litres
- (F) 2,55,000 litres
- (G) 3,00,000 litres

53. Pseudopodia ('false feet') help the amoeba both move and capture food. If an amoeba moves 0.2 mm per minute, how far does it move in 1 hour? [\[Nutrition in Animals\]](#)

- (A) 60 mm
- (B) 120 mm
- (C) 12 mm
- (D) 6 mm
- (E) 1.2 mm
- (F) 0.2 mm
- (G) 20 mm

54. A triangle has angles $3y$, $4y$ and $5y$. After finding y , by how much does the largest angle exceed the smallest? [\[The Triangle & its Properties\]](#)

- (A) 75 deg
- (B) 30 deg
- (C) 15 deg
- (D) 90 deg
- (E) 45 deg
- (F) 20 deg
- (G) 60 deg

55. Light bounces off a smooth mirror in a regular way (regular reflection) but scatters off a rough wall (diffuse reflection). If 100 parallel rays hit a mirror, how many stay parallel after reflecting? [\[Light\]](#)

- (A) 1
- (B) 99
- (C) 10
- (D) 25
- (E) 50
- (F) 100 (all stay parallel)
- (G) 0

56. Round each to the nearest thousand and add: $12,640 + 8,380 + 5,910$. [\[Large Numbers\]](#)

- (A) 27,930
- (B) 28,000
- (C) 26,000
- (D) 26,930
- (E) 30,000
- (F) 27,000
- (G) 25,000

57. Starch from food is finally broken into glucose, which the blood carries to cells. If 1 g of glucose gives about 16 kJ of energy and a meal yields 45 g of glucose, how much energy can it provide? [\[Nutrition in Animals\]](#)

- (A) 700 kJ
- (B) 16 kJ
- (C) 360 kJ
- (D) 720 kJ
- (E) 29 kJ
- (F) 450 kJ
- (G) 61 kJ

58. Sides of a triangle are 13 cm, 13 cm and 10 cm. The altitude to the 10 cm base splits it in half. What is the height (altitude)? [\[The Triangle & its Properties\]](#)

- (A) 6 cm
- (B) 8 cm
- (C) 144 cm
- (D) 5 cm
- (E) 12 cm
- (F) 10 cm
- (G) 23 cm

59. A torch beam hits a mirror and the reflected beam makes a 70 deg angle with the mirror surface. Light reflection is measured from the normal (perpendicular). What is the angle of reflection (from the normal)?

[Light]

- (A) 20 deg
- (B) 35 deg
- (C) 110 deg
- (D) 40 deg
- (E) 90 deg
- (F) 70 deg
- (G) 50 deg

60. The population of a city was 8,45,000 and grew by 1,28,000 over five years. What is the new population, and roughly what is it to the nearest lakh? [Large Numbers]

- (A) 9,73,000; about 10 lakh
- (B) 9,73,000; about 9 lakh
- (C) 9,73,000; about 9.5 lakh
- (D) 9,37,000; about 9 lakh
- (E) 10,73,000; about 11 lakh
- (F) 7,17,000; about 7 lakh
- (G) 9,73,000; exactly 9 lakh

61. Ruminants house helpful bacteria that digest cellulose for them. If these bacteria break down 18 g of cellulose per hour in one cow, how much do they break down in a 24-hour day across 5 cows (same rate)?

[Nutrition in Animals]

- (A) 90 g
- (B) 2160 g
- (C) 1080 g
- (D) 2106 g
- (E) 120 g
- (F) 216 g
- (G) 432 g

62. Can a triangle have sides 4 cm, 6 cm and 11 cm? Explain using the triangle inequality. [The Triangle & its Properties]

- (A) No, because $4 + 6 = 10$ is less than 11
- (B) No, because $4 + 6 = 10$ is more than 11
- (C) No, because all sides must be equal
- (D) Yes, because the sides are whole numbers
- (E) Yes, all sides are different
- (F) Yes, because $6 + 11$ is more than 4

63. A spherical mirror's radius of curvature is twice its focal length. If a concave mirror has a focal length of 15 cm, what is its radius of curvature? [Light]

- (A) 15 cm
- (B) 17 cm
- (C) 45 cm
- (D) 30 cm
- (E) 20 cm
- (F) 7.5 cm

64. A printing press prints 1,250 pages per minute. How many pages does it print in 1 hour 12 minutes?

[Large Numbers]

- (A) 1,00,000 pages
- (B) 15,000 pages
- (C) 75,000 pages
- (D) 84,000 pages
- (E) 90,000 pages
- (F) 9,000 pages
- (G) 1,500 pages

65. Food moves through the food pipe by waves of muscle squeezing called peristalsis, not by gravity. If one squeeze pushes food 4 cm and a wave makes 6 squeezes, how far does food travel along a 30 cm pipe, and does it reach the end? [Nutrition in Animals]

- (A) 24 cm, overshooting the end
- (B) 30 cm, reaching the end exactly
- (C) 24 cm, so it does NOT yet reach the end
- (D) 26 cm, reaching the end
- (E) 24 cm, so it just reaches the end
- (F) 20 cm, reaching the end
- (G) 10 cm, not reaching the end

66. A set square has a 90 deg corner and its other two corners are equal to each other. Work out how many degrees each of those two equal corners measures. [The Triangle & its Properties]

- (A) 45.5 deg
- (B) 45 deg
- (C) 30 deg
- (D) 60 deg
- (E) 90 deg
- (F) 50 deg
- (G) 40 deg

67. Through a pinhole, an image of a 2 m tall tree forms 4 cm tall when the screen is 20 cm from the hole and the tree is 10 m away. If the screen is moved to 40 cm, the image height doubles. What is the new image height? [Light]

- (A) 10 cm
- (B) 2 cm
- (C) 16 cm
- (D) 6 cm
- (E) 12 cm
- (F) 4 cm
- (G) 8 cm

68. Subtract: 10,00,000 - 4,56,789. [Large Numbers]

- (A) 4,56,789
- (B) 5,43,211
- (C) 5,44,211
- (D) 6,43,211
- (E) 5,53,211
- (F) 5,43,221

69. Hydrochloric acid in the stomach kills germs and helps enzymes work. If the stomach makes 2 L of gastric juice a day and acid is $\frac{1}{250}$ of it by volume, how many millilitres of acid is that? [\[Nutrition in Animals\]](#)

- (A) 2 mL
- (B) 16 mL
- (C) 20 mL
- (D) 250 mL
- (E) 8 mL
- (F) 4 mL

70. The perimeter of an isosceles triangle is 50 cm and its base is 16 cm. What is the length of each equal side? [\[The Triangle & its Properties\]](#)

- (A) 25 cm
- (B) 17 cm
- (C) 33 cm
- (D) 16 cm
- (E) 8 cm
- (F) 34 cm

71. In a periscope, two mirrors are set at 45 deg to send light around a corner, turning it through two right angles in total. Through how many degrees is the light's path turned altogether? [\[Light\]](#)

- (A) 180 deg
- (B) 270 deg
- (C) 45 deg twice with no change
- (D) 360 deg
- (E) 90 deg
- (F) 135 deg
- (G) 45 deg

72. Using digits 3, 0, 5, 8 each once, what is the largest 4-digit number you can form, and the smallest (no leading zero)? [\[Large Numbers\]](#)

- (A) Largest 8530, smallest 3085
- (B) Largest 8350, smallest 3058
- (C) Largest 8530, smallest 3508
- (D) Largest 8503, smallest 3058
- (E) Largest 8530, smallest 3058
- (F) Largest 5830, smallest 3058

73. A hungry python swallows prey whole and digests it slowly over days. If it digests $\frac{1}{8}$ of a 4 kg meal each day, how many grams remain undigested after 3 days? [\[Nutrition in Animals\]](#)

- (A) 2500 g
- (B) 500 g
- (C) 2000 g
- (D) 3000 g
- (E) 1500 g
- (F) 1000 g
- (G) 3500 g

74. In a right triangle, one leg is 5 cm and the hypotenuse is 13 cm. What is the length of the other leg? [\[The](#)

[Triangle & its Properties\]](#)

- (A) 6.5 cm
- (B) 8 cm
- (C) 144 cm
- (D) 9 cm
- (E) 12 cm
- (F) 18 cm
- (G) 14 cm

75. A candle flame and its mirror image are 64 cm apart. The mirror is exactly halfway between them. How far is the candle from the mirror? [\[Light\]](#)

- (A) 32 cm
- (B) 64 cm
- (C) 30 cm
- (D) 128 cm
- (E) 8 cm
- (F) 48 cm
- (G) 16 cm

76. A company earns 2,75,000 rupees a week. Estimate its yearly earnings (52 weeks) by rounding the weekly amount to the nearest lakh. [\[Large Numbers\]](#)

- (A) 1,43,00,000 rupees
- (B) 2,60,00,000 rupees
- (C) 52,00,000 rupees
- (D) 1,30,00,000 rupees
- (E) 1,56,00,000 rupees
- (F) 3,00,000 rupees
- (G) 1,56,000 rupees

77. Nutrition that takes ready-made food from another living host is called parasitic. If a tapeworm in the gut absorbs 2% of the 2500 kJ a person eats daily, how much energy does the person lose to it in a week?

[\[Nutrition in Animals\]](#)

- (A) 700 kJ
- (B) 350 kJ
- (C) 175 kJ
- (D) 50 kJ
- (E) 250 kJ
- (F) 2450 kJ

78. Two interior angles of a triangle add up to 110 deg. What is the exterior angle at the third vertex? [\[The](#)

[Triangle & its Properties\]](#)

- (A) 110 deg
- (B) 90 deg
- (C) 250 deg
- (D) 35 deg
- (E) 55 deg
- (F) 180 deg
- (G) 70 deg

- 79.** Of the seven spectrum colours, red bends least and violet bends most through a prism. If a list orders them by bending, which position (1st = least) does green occupy among R, O, Y, G, B, I, V? [\[Light\]](#)
- (A) 1st
 - (B) 6th
 - (C) 2nd
 - (D) 5th
 - (E) 4th
 - (F) 3rd
- 80.** How many lakhs make 1 crore, and how many crores make 100 crore? [\[Large Numbers\]](#)
- (A) 10 lakhs make a crore; 10 crore make 100 crore
 - (B) 10 lakhs make a crore; 100 crore make 100 crore
 - (C) 100 lakhs make a crore; 10 crore make 100 crore
 - (D) 1000 lakhs make a crore; 100 crore make 100 crore
 - (E) 100 lakhs make a crore; 1 crore makes 100 crore
 - (F) 100 lakhs make a crore; 100 crore make 100 crore
- 81.** Different teeth do different jobs: incisors bite, canines tear, premolars and molars grind. If a child has 20 milk teeth and 8 of them are molars-type grinders, what fraction are NOT grinders? [\[Nutrition in Animals\]](#)
- (A) $\frac{1}{2}$
 - (B) $\frac{5}{3}$
 - (C) $\frac{2}{5}$
 - (D) $\frac{3}{4}$
 - (E) $\frac{8}{20}$
 - (F) $\frac{3}{5}$
- 82.** A flagpole casts no shadow problem: a triangle has a base of 20 cm and a height of 9 cm. A second triangle has the same area but a base of 12 cm. What is its height? [\[The Triangle & its Properties\]](#)
- (A) 10 cm
 - (B) 15 cm
 - (C) 18 cm
 - (D) 5.4 cm
 - (E) 7.5 cm
 - (F) 30 cm
- 83.** A laser pointer shines a spot on a screen 5 m away. If you tilt the pointer so the spot moves 0.6 m sideways, and a second screen at 10 m would move the spot proportionally, how far does the spot move on the farther screen? [\[Light\]](#)
- (A) 0.3 m
 - (B) 0.06 m
 - (C) 2.4 m
 - (D) 1.0 m
 - (E) 6 m
 - (F) 1.2 m
 - (G) 0.6 m

84. A satellite orbits Earth at 28,000 km per hour. How far does it travel in 3 hours 30 minutes? [\[Large Numbers\]](#)

- (A) 9,800 km
- (B) 98,000 km
- (C) 98,00,000 km
- (D) 8,400 km
- (E) 1,12,000 km
- (F) 94,000 km
- (G) 84,000 km

85. Bile does not contain enzymes but breaks large fat drops into tiny droplets so enzymes work faster. If breaking one big drop makes 1000 tiny droplets, how many tiny droplets come from 25 big drops? [\[Nutrition in Animals\]](#)

- (A) 5000 droplets
- (B) 2500 droplets
- (C) 25000 droplets
- (D) 40 droplets
- (E) 250 droplets
- (F) 100000 droplets

86. In triangle PQR, angle P is twice angle Q, and angle R is 60 deg. What is angle Q? [\[The Triangle & its Properties\]](#)

- (A) 60 deg
- (B) 50 deg
- (C) 45 deg
- (D) 40 deg
- (E) 30 deg
- (F) 80 deg

87. The image in a plane mirror is the same size as the object (magnification 1). If a 15 cm pencil and its image are viewed, what is the COMBINED length of pencil plus image? [\[Light\]](#)

- (A) 45 cm
- (B) 7.5 cm
- (C) 30 cm
- (D) 20 cm
- (E) 1 cm
- (F) 15 cm

88. Find the value: $9,00,000 + 9 \times 10,000 + 9 \times 100 + 9$. [\[Large Numbers\]](#)

- (A) 9,09,909
- (B) 9,90,099
- (C) 9,99,999
- (D) 9,00,909
- (E) 9,90,999
- (F) 9,90,909
- (G) 9,91,909

89. Absorbed food is carried by blood to all cells, where it is broken down to release energy (this is also why we breathe). If a cell uses 6 units of glucose and 6 units of oxygen to make energy, how many oxygen units are needed for 30 glucose units? [\[Nutrition in Animals\]](#)

- (A) 180 units
- (B) 15 units
- (C) 36 units
- (D) 6 units
- (E) 60 units
- (F) 30 units
- (G) 5 units

90. A square has a side of 10 cm. A diagonal divides it into two right triangles. What is the area of ONE such triangle? [\[The Triangle & its Properties\]](#)

- (A) 50 square cm
- (B) 10 square cm
- (C) 40 square cm
- (D) 200 square cm
- (E) 20 square cm
- (F) 25 square cm

91. A spotlight makes a circular bright patch 2 m wide on a wall 4 m away. Because light travels straight, the patch widens with distance. How wide is the patch on a wall 12 m away (same beam)? [\[Light\]](#)

- (A) 2 m
- (B) 8 m
- (C) 6 m
- (D) 3 m
- (E) 24 m
- (F) 4 m

92. A truck carries 18 boxes, each holding 2,500 screws. A warehouse receives 40 such trucks. How many screws is that in total? [\[Large Numbers\]](#)

- (A) 18,000 screws
- (B) 1,000 screws
- (C) 1,80,000 screws
- (D) 1,80,00,000 screws
- (E) 45,000 screws
- (F) 18,00,000 screws
- (G) 7,20,000 screws

93. The food pipe (oesophagus) and windpipe both open at the back of the mouth; a flap stops food entering the windpipe. If swallowing takes 2 seconds and you swallow about 600 times a day, how many minutes a day are spent swallowing? [\[Nutrition in Animals\]](#)

- (A) 1200 minutes
- (B) 30 minutes
- (C) 10 minutes
- (D) 60 minutes
- (E) 5 minutes
- (F) 12 minutes
- (G) 20 minutes

94. The three angles of a triangle are $(x + 10)$, $(2x)$ and $(3x - 10)$ degrees. What is the value of x ? [\[The Triangle](#)

[& its Properties\]](#)

- (A) 40
- (B) 18
- (C) 60
- (D) 20
- (E) 36
- (F) 30
- (G) 25

95. Sunlight reflecting off a still pond shows a clear image, but a rippled pond does not. If a smooth surface reflects 95% of light regularly and a rough one only 5% regularly, how many times more regular reflection does the smooth surface give? [\[Light\]](#)

- (A) 15 times
- (B) 100 times
- (C) 19 times
- (D) 9.5 times
- (E) 90 times
- (F) 5 times
- (G) 2 times

96. Round 3,499 and 4,501 each to the nearest thousand, then find the difference of the rounded values.

[\[Large Numbers\]](#)

- (A) 8,000
- (B) 0
- (C) 3,000
- (D) 1,002
- (E) 1,000
- (F) 1,500
- (G) 2,000

97. Energy from food keeps the body warm and active. If a 12-year-old needs about 2100 kJ for rest and $\frac{1}{3}$ more than that for active play, what is the total daily need? [\[Nutrition in Animals\]](#)

- (A) 3150 kJ
- (B) 700 kJ
- (C) 2800 kJ
- (D) 1400 kJ
- (E) 2400 kJ
- (F) 6300 kJ
- (G) 2200 kJ

98. A rectangular field is 40 m by 30 m. A path runs along its diagonal. How long is the diagonal path? [\[The](#)

[Triangle & its Properties\]](#)

- (A) 10 m
- (B) 50 m
- (C) 70 m
- (D) 60 m
- (E) 35 m
- (F) 2500 m

99. An object 8 cm tall stands before a concave mirror that gives an image 3 times taller and inverted. What is the size of the image, and how does it compare to the object? [\[Light\]](#)

- (A) 11 cm, larger
- (B) 24 cm, the same size
- (C) 16 cm, twice as large
- (D) 24 cm, three times larger
- (E) 5 cm, smaller
- (F) 8 cm, same size
- (G) 2.67 cm, smaller

100. The distance from Earth to the Moon is about 3,84,000 km. A spacecraft has covered 2,96,500 km. How much farther must it travel, to the nearest thousand? [\[Large Numbers\]](#)

- (A) 80,000 km
- (B) 90,000 km
- (C) 6,80,500 km
- (D) 88,000 km
- (E) 87,500 km
- (F) 1,08,000 km
- (G) 87,000 km